

WEEK 2 PROGRESS REPORT

WHO DEFINES THE PROBLEM? A COMPUTATIONAL ANALYSIS OF HOW EU INSTITUTIONS FRAME ARTIFICIAL INTELLIGENCE REGULATION

4 September 2026

Project GitHub Link: <https://github.com/michelle-qyb/SOSC314-Course-Project/>

Division of Responsibilities

For this project stage, each group member was responsible for the following components:

Tim Maurice Steiner:

Primary responsibility: Research design and substantive framing — scope decisions, conceptual framework, written deliverables.

Specific tasks completed this week: Defined the inclusion/exclusion rules and document-type classification (doc_types.csv); wrote the query code (cellar.py) and inventory code (01_inventory.py); reviewed the inventory results and examples; wrote and edited the progress report and the figures; set up the GitHub repository structure and README.

Yuhan Qiu:

Primary responsibility: Data infrastructure and computational pipeline — API access, corpus construction, reproducibility.

Specific tasks completed this week: Tested CELLAR (SPARQL + REST) as the data source; wrote the feasibility and exploration code (01_feasibility_test_and_initial_exploration.py); cross-checked the inventory counts and identified the jointly authored works issue.

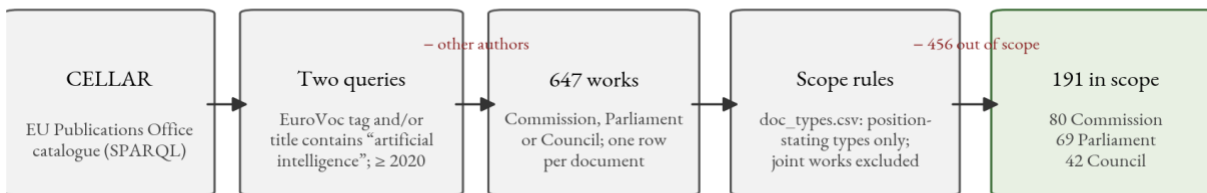
Research Question

How do the three EU legislative institutions — the European Commission, the European Parliament and the Council of the EU — frame artificial intelligence as a policy problem, and does institutional position predict the frames an institution uses? The design holds the regulatory environment constant (one polity, one legislative file: the AI Act of 2021–24 and its 2025–26 revision) and varies institutional role: executive initiator, elected legislature, and member-state governments.

Data Source

The corpus is API-based. Documents from all three institutions are retrievable from CELLAR, the EU Publications Office repository behind EUR-Lex. CELLAR exposes a public SPARQL endpoint for metadata and a REST interface for full text; neither requires registration and nothing is scraped. Metadata queries return identifier, authoring body, resource type, date and title; an HTTP request on the identifier returns the English full text. Retrieval was verified end to end on the 2020 White Paper on AI.

Two retrieval criteria are combined: works indexed with the EuroVoc subject “artificial intelligence”, and works whose English title contains the phrase. The second matters for the Council: its register documents are mirrored in CELLAR but rarely EuroVoc-tagged, so a tag-only query returned 30 Council works, half of them co-signed legal acts, while the title query recovered over 90. Figure 1 summarises the acquisition pipeline from catalogue query to in-scope corpus.



Full texts are retrieved from the same identifiers via CELLAR REST (verified on the 2020 White Paper); collection of all 191 texts follows in Week 3.

Figure 1. Data-acquisition and scope pipeline: from the CELLAR catalogue to the 191-document corpus (retrieval of 4 September 2026).

Unit of Analysis

The document is the unit of collection and of institution-level comparison. Frames will be measured at paragraph level and aggregated to documents, because institutional texts are long and mix several frames. Institution is the explanatory variable; document type and legislative stage are controls.

Initial Exploration

The queries returned 647 distinct works authored by the three institutions since January 2020 (retrieval of 4 September 2026; CELLAR is continuously indexed, so counts drift slightly between days and the committed snapshot in the repository is the reference). Applying the scope rules below leaves 191 in-scope documents: 80 Commission, 69 Parliament, 42 Council (Figure 2). Three patterns stand out. First, output tracks the legislative calendar: documents surge around major legislative developments — the AI Act proposal (April 2021) and the Digital Omnibus on AI (November 2025) — while 2023, the trilogue year, is quietest because negotiation happens in restricted documents. Second, the institutions produce different genres: the Commission mainly staff working documents and communications; Parliament resolutions and committee opinions; the Council notes, conclusions and positions. This makes document type an important consideration for the subsequent text analysis. Finally, representative documents already suggest substantively distinct institutional emphases, from the Commission’s 2020 White Paper on AI, to Parliament’s 2020 resolution on the ethical aspects of AI, to the Council’s 2022 General Approach on the AI Act. Overall, the corpus is sufficiently large and diverse for comparative analysis.

Scope Decisions

All inclusion and exclusion rules derive from a single test: is the text the institution itself stating or negotiating a policy position? The rules were built inductively — every document type returned by the queries was inspected through example documents and classified, with a written reason per type, in data/doc_types.csv.

Included: document types in which an institution states or negotiates a position — communications, white papers, staff working documents, impact assessments, proposals, recommendations and reports; EP resolutions, adopted texts, committee reports, opinions and amendment lists; Council conclusions, positions, presidency notes and item notes.

Excluded: 334 general publications (JRC and DG research reports and briefs), which are research output rather than institutional positions and form a labelled secondary corpus; adopted legal acts, which are co-signed and identical across institutions; Council cover notes and register copies of Commission proposals, which the Council forwarded but did not write; merger notices tagged AI, plenary minutes, factsheets and corrigenda; and works attributed to more than one institution (co-signed final acts and joint declarations), which cannot serve

an institutional comparison. Observation window: 1 January 2020 — chosen because the February 2020 White Paper opens the current policy cycle — to present. Language: English.

Feasibility Assessment

Feasible. Access is programmatic and does not require authentication; metadata can be retrieved reproducibly in under a minute, and full-text retrieval was verified end to end on the 2020 White Paper. Data quality is high: core metadata (identifier, institution, date, type) is complete for all records; English titles are missing for 13 records (2%), a minor gap that will be monitored. Three risks shape the analysis rather than block it. (1) Genre confound: institutional differences are entangled with document type, so models will control for type and key comparisons will be run within genre. (2) Textual inheritance: Council compromise texts and EP amendment lists may reproduce the Commission's base text rather than reflect independent framing. Works co-signed by multiple institutions were already excluded at the corpus-construction stage; for the remaining negotiation documents, an inherited-share variable will quantify this and non-inherited passages will be analysed separately. (3) Institutional imbalance: the in-scope corpus is uneven across institutions, with the Council notably smaller (42 vs. 69–80) and concentrated in short, procedural document types. This limits the depth of Council-specific findings, which will carry this caveat rather than be treated as fully comparable in scale to the other two institutions.

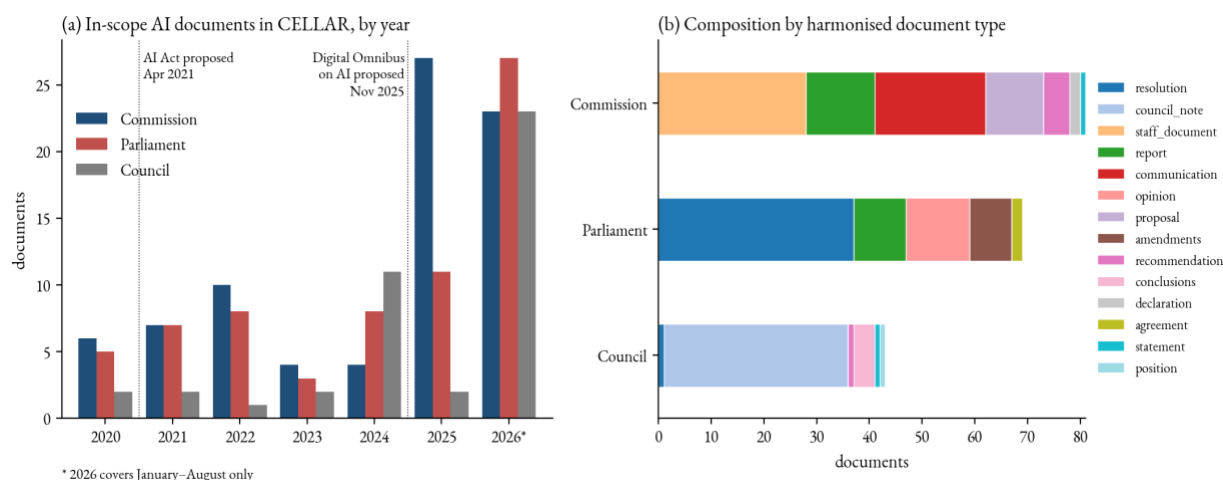


Figure 2. Inventory of AI-related institutional documents retrievable from CELLAR ($n = 191$ in scope of 647 found; retrieval 4 September 2026). Panel (a): in-scope documents by year and institution. Panel (b): composition by harmonised document type.